PRAKTIKUM 1

JARINGAN TELEKOMUNIKASI DAN INFORMASI

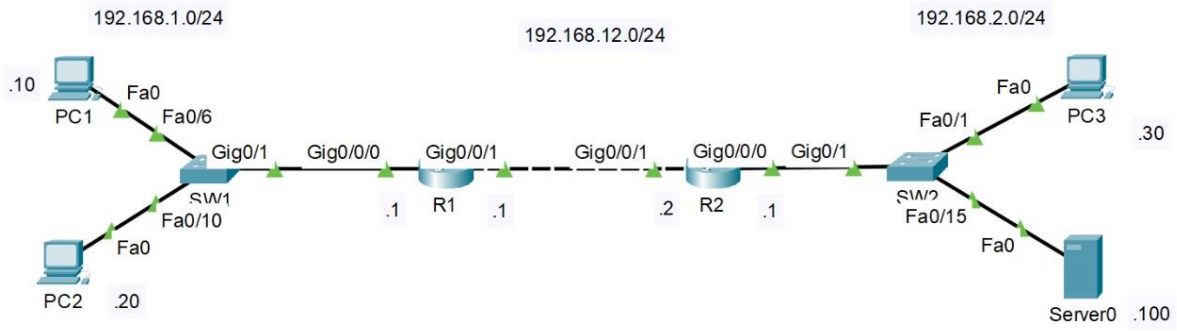
Nama :

- Devita Syifa Azzahraa (221331009)
- Ghania Fitriasy Syifa (221331013)

Kelas : 2A TC

Tanggal Praktikum : 26/10/2023

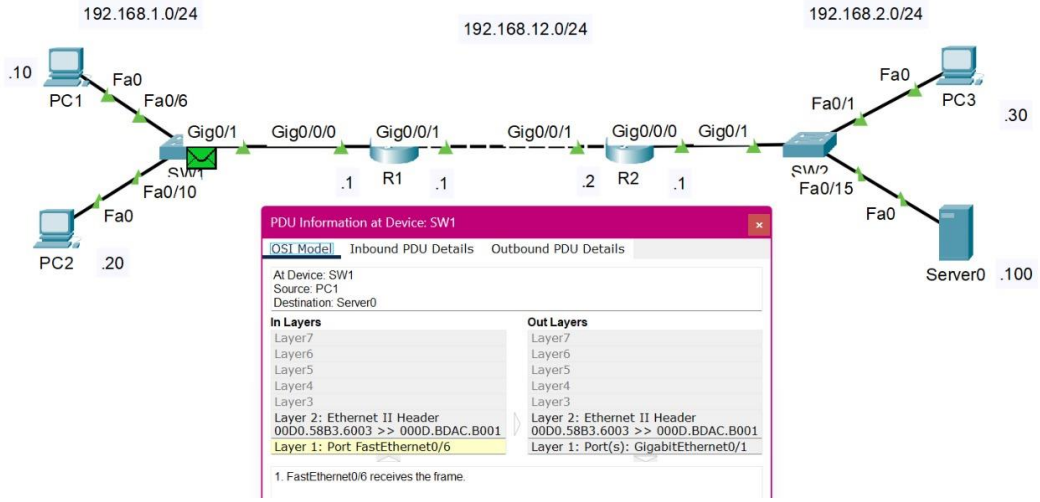
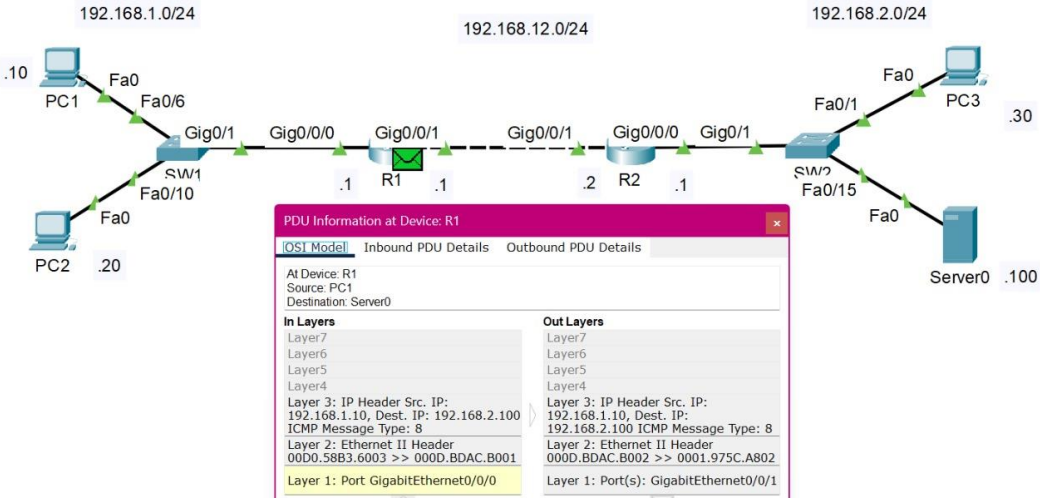
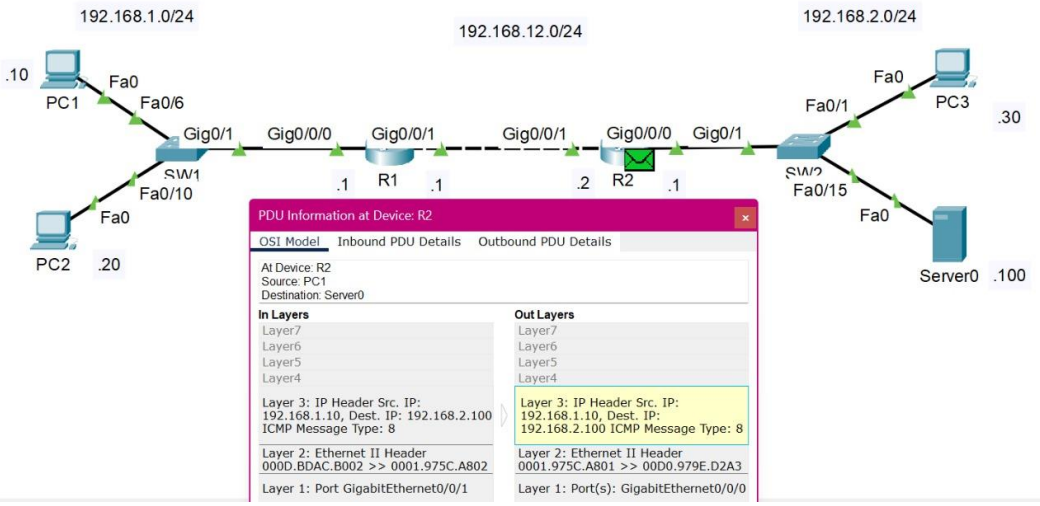
A. GAMBAR RANGKAIAN

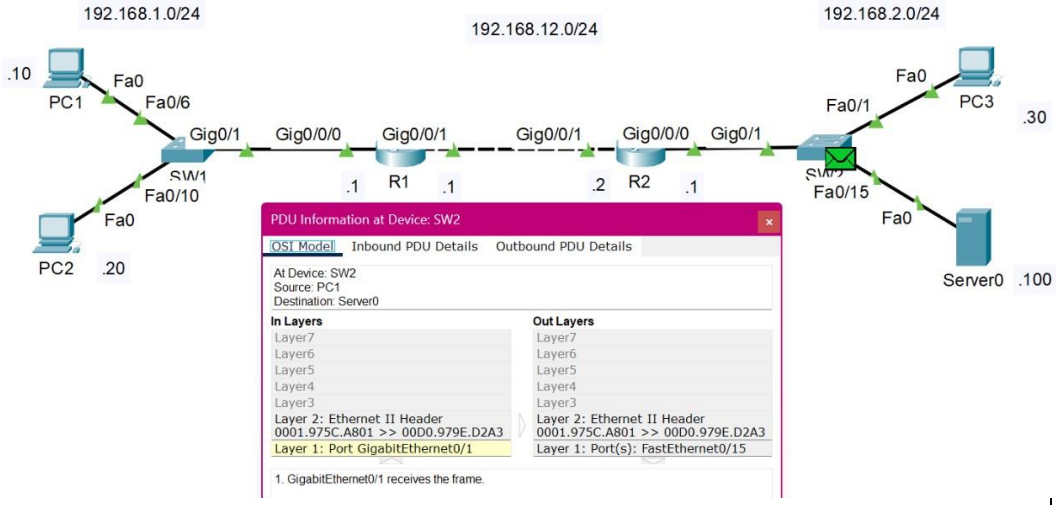
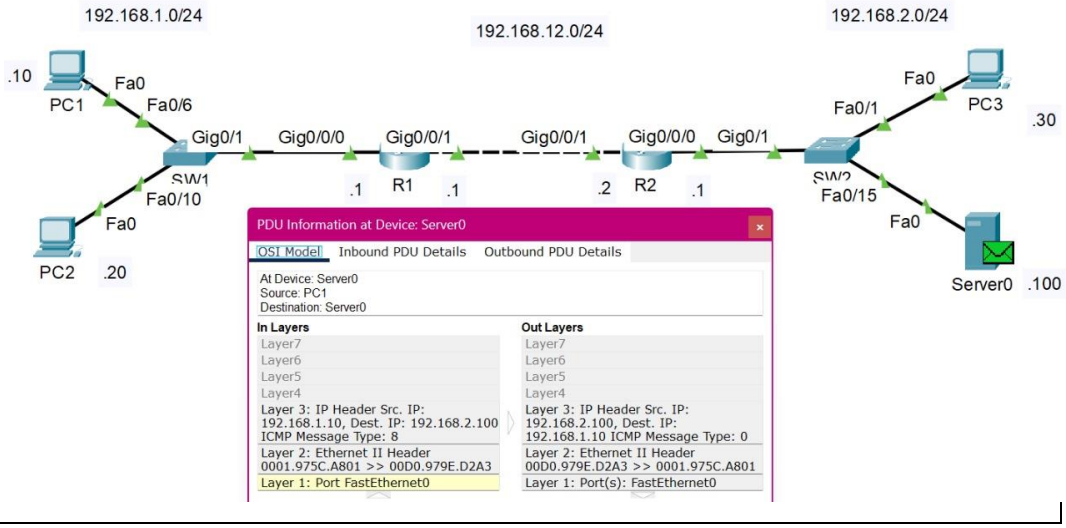


B. HASIL PRAKTIKUM

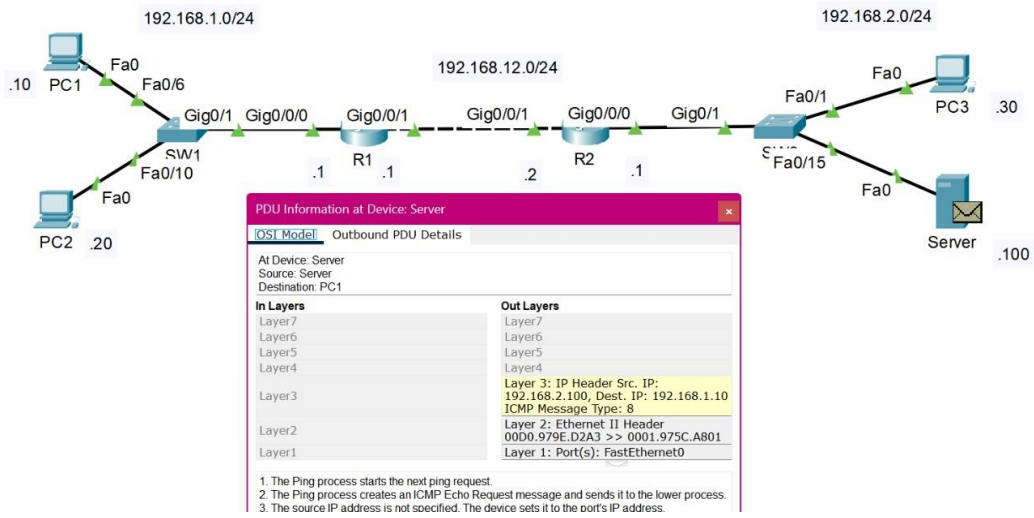
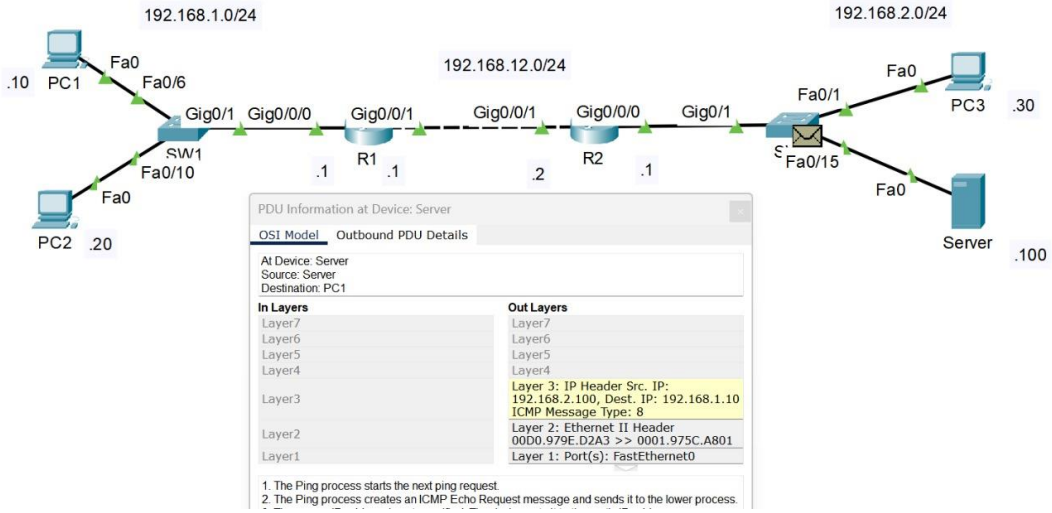
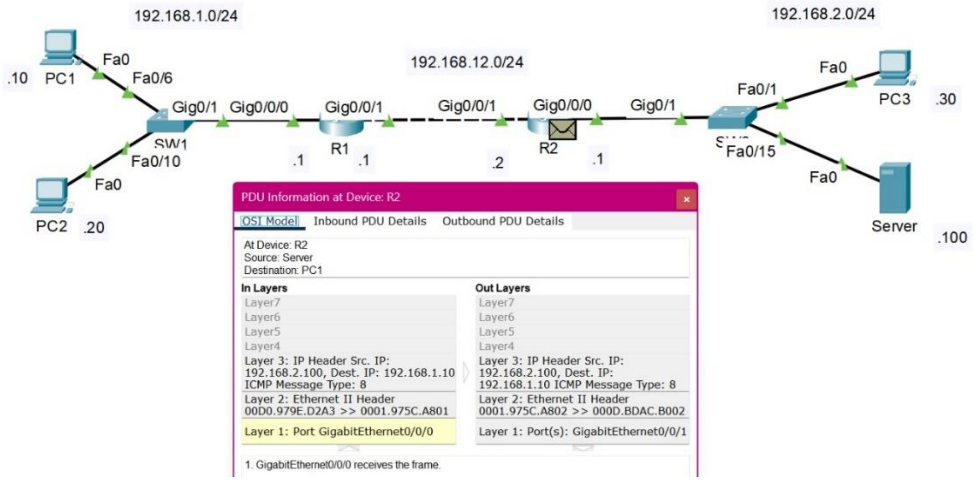
1. PC1 – SW1 – R1 – R2 – SW2 – Server

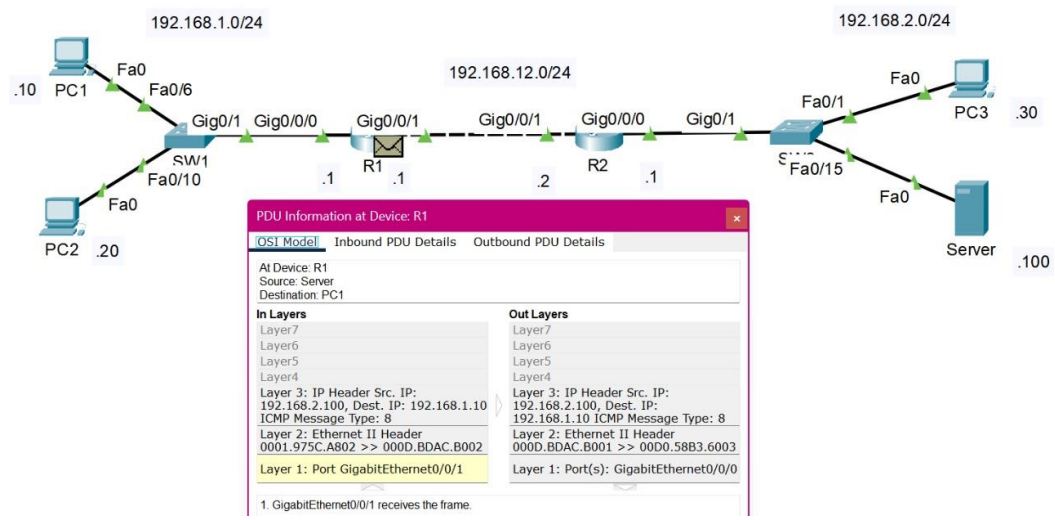
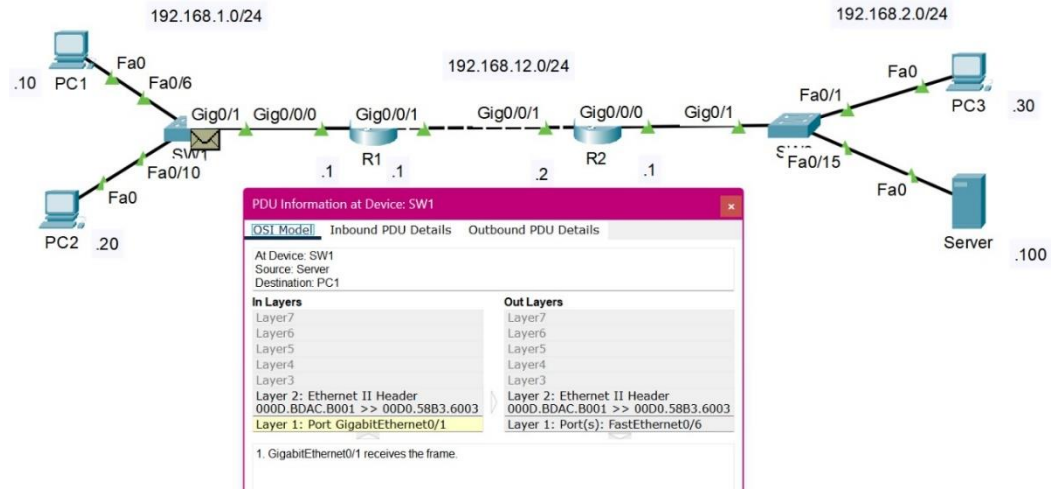
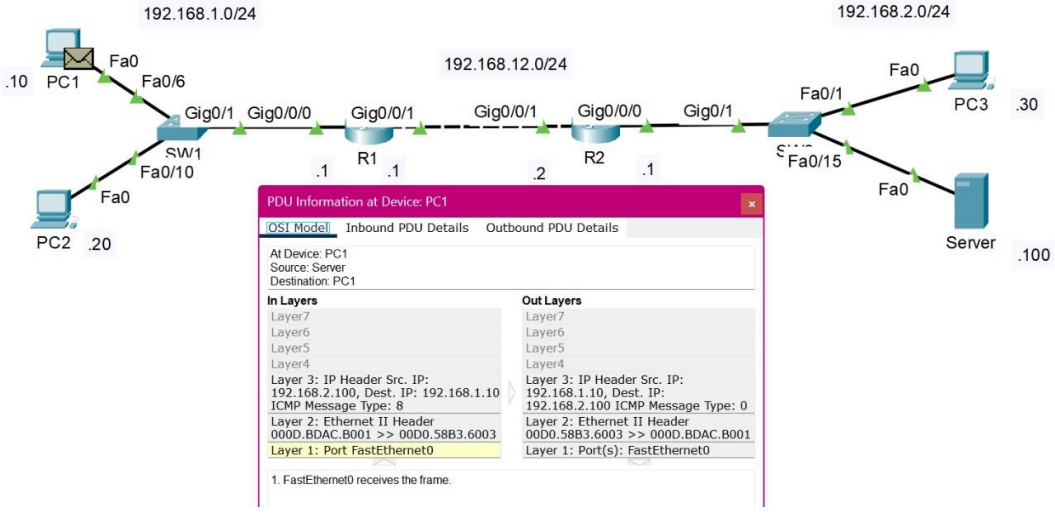
Network-End Device	Keterangan
PC1	

SW1	 192.168.1.0/24 192.168.12.0/24 192.168.2.0/24 .10 PC1 Fa0 Fa0/6 Gig0/1 Gig0/0/0 Gig0/0/1 Gig0/0/1 Gig0/0/0 Gig0/1 SW1 Fa0/10 Fa0 PC2 .20 .1 R1 .1 .2 R2 .1 SW2 Fa0/1 Fa0 PC3 .30 Fa0/15 Fa0 Server0 .100 PDU Information at Device: SW1 OSI Model Inbound PDU Details Outbound PDU Details At Device: SW1 Source: PC1 Destination: Server0 In Layers Out Layers Layer7 Layer6 Layer5 Layer4 Layer3 Layer 2: Ethernet II Header 00D0.58B3.6003 >> 000D.BDAC.B001 Layer 1: Port FastEthernet0/6 Layer 2: Ethernet II Header 00D0.58B3.6003 >> 000D.BDAC.B001 Layer 1: Port(s): GigabitEthernet0/1 1. FastEthernet0/6 receives the frame.
R1	 192.168.1.0/24 192.168.12.0/24 192.168.2.0/24 .10 PC1 Fa0 Fa0/6 Gig0/1 Gig0/0/0 Gig0/0/1 Gig0/0/1 Gig0/0/0 Gig0/1 SW1 Fa0/10 Fa0 PC2 .20 .1 R1 .1 .2 R2 .1 SW2 Fa0/1 Fa0 PC3 .30 Fa0/15 Fa0 Server0 .100 PDU Information at Device: R1 OSI Model Inbound PDU Details Outbound PDU Details At Device: R1 Source: PC1 Destination: Server0 In Layers Out Layers Layer7 Layer6 Layer5 Layer4 Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 192.168.2.100 ICMP Message Type: 8 Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 192.168.2.100 ICMP Message Type: 8 Layer 2: Ethernet II Header 00D0.58B3.6003 >> 000D.BDAC.B001 Layer 2: Ethernet II Header 000D.BDAC.B002 >> 0001.975C.A802 Layer 1: Port GigabitEthernet0/0/0 Layer 1: Port(s): GigabitEthernet0/0/1
R2	 192.168.1.0/24 192.168.12.0/24 192.168.2.0/24 .10 PC1 Fa0 Fa0/6 Gig0/1 Gig0/0/0 Gig0/0/1 Gig0/0/1 Gig0/0/0 Gig0/1 SW1 Fa0/10 Fa0 PC2 .20 .1 R1 .1 .2 R2 .1 SW2 Fa0/1 Fa0 PC3 .30 Fa0/15 Fa0 Server0 .100 PDU Information at Device: R2 OSI Model Inbound PDU Details Outbound PDU Details At Device: R2 Source: PC1 Destination: Server0 In Layers Out Layers Layer7 Layer6 Layer5 Layer4 Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 192.168.2.100 ICMP Message Type: 8 Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 192.168.2.100 ICMP Message Type: 8 Layer 2: Ethernet II Header 000D.BDAC.B002 >> 0001.975C.A802 Layer 2: Ethernet II Header 0001.975C.A801 >> 00D0.979E.D2A3 Layer 1: Port GigabitEthernet0/0/1 Layer 1: Port(s): GigabitEthernet0/0/0

SW2	 192.168.1.0/24 192.168.12.0/24 192.168.2.0/24 .10 PC1 Fa0 Fa0/6 Gig0/1 Gig0/0/0 Gig0/0/1 Gig0/0/1 Gig0/0/0 Gig0/1 SW1 Fa0/10 Fa0 PC2 .20 .1 R1 .1 .2 R2 .1 Fa0/1 Fa0 PC3 .30 Fa0/15 Fa0 Server0 .100 PDU Information at Device: SW2 OSI Model Inbound PDU Details Outbound PDU Details At Device: SW2 Source: PC1 Destination: Server0 In Layers Out Layers Layer7 Layer6 Layer5 Layer4 Layer3 Layer 2: Ethernet II Header 0001.975C.A801 >> 00D0.979E.D2A3 Layer 2: Ethernet II Header 0001.975C.A801 >> 00D0.979E.D2A3 Layer 1: Port GigabitEthernet0/1 Layer 1: Port(s): FastEthernet0/15 1. GigabitEthernet0/1 receives the frame.
Server	 192.168.1.0/24 192.168.12.0/24 192.168.2.0/24 .10 PC1 Fa0 Fa0/6 Gig0/1 Gig0/0/0 Gig0/0/1 Gig0/0/1 Gig0/0/0 Gig0/1 SW1 Fa0/10 Fa0 PC2 .20 .1 R1 .1 .2 R2 .1 Fa0/1 Fa0 PC3 .30 Fa0/15 Fa0 Server0 .100 PDU Information at Device: Server0 OSI Model Inbound PDU Details Outbound PDU Details At Device: Server0 Source: PC1 Destination: Server0 In Layers Out Layers Layer7 Layer6 Layer5 Layer4 Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 192.168.2.100 ICMP Message Type: 8 Layer 3: IP Header Src. IP: 192.168.2.100, Dest. IP: 192.168.1.10 ICMP Message Type: 0 Layer 2: Ethernet II Header 0001.975C.A801 >> 00D0.979E.D2A3 Layer 2: Ethernet II Header 00D0.979E.D2A3 >> 0001.975C.A801 Layer 1: Port FastEthernet0 Layer 1: Port(s): FastEthernet0

2. Server – SW2 – R2 – R1 – SW1 – PC1

Network-End Device	Keterangan
Server	
SW2	
R2	

R1	 192.168.1.0/24 192.168.2.0/24 192.168.12.0/24 .10 PC1 Fa0 Fa0/6 Gig0/1 Gig0/0/0 Gig0/0/1 Gig0/0/1 Gig0/0/0 Gig0/1 SW1 Fa0/10 Fa0 PC2 .20 .1 R1 .1 .2 R2 .1 Fa0/1 Fa0/15 Fa0 PC3 .30 Fa0 Server .100 PDU Information at Device: R1 OSI Model Inbound PDU Details Outbound PDU Details At Device: R1 Source: Server Destination: PC1 In Layers Layer7 Layer6 Layer5 Layer4 Layer 3: IP Header Src. IP: 192.168.2.100, Dest. IP: 192.168.1.10 ICMP Message Type: 8 Layer 2: Ethernet II Header 0001.975C.A802 >> 000D.BDAC.B002 Layer 1: Port GigabitEthernet0/0/1 Out Layers Layer7 Layer6 Layer5 Layer4 Layer 3: IP Header Src. IP: 192.168.2.100, Dest. IP: 192.168.1.10 ICMP Message Type: 8 Layer 2: Ethernet II Header 000D.BDAC.B001 >> 00D0.58B3.6003 Layer 1: Port(s): GigabitEthernet0/0/0 1. GigabitEthernet0/0/1 receives the frame.
SW1	 192.168.1.0/24 192.168.2.0/24 192.168.12.0/24 .10 PC1 Fa0 Fa0/6 Gig0/1 Gig0/0/0 Gig0/0/1 Gig0/0/1 Gig0/0/0 Gig0/1 SW1 Fa0/10 Fa0 PC2 .20 .1 R1 .1 .2 R2 .1 Fa0/1 Fa0/15 Fa0 PC3 .30 Fa0 Server .100 PDU Information at Device: SW1 OSI Model Inbound PDU Details Outbound PDU Details At Device: SW1 Source: Server Destination: PC1 In Layers Layer7 Layer6 Layer5 Layer4 Layer3 Layer 2: Ethernet II Header 000D.BDAC.B001 >> 00D0.58B3.6003 Layer 1: Port GigabitEthernet0/1 Out Layers Layer7 Layer6 Layer5 Layer4 Layer3 Layer 2: Ethernet II Header 000D.BDAC.B001 >> 00D0.58B3.6003 Layer 1: Port(s): FastEthernet0/6 1. GigabitEthernet0/1 receives the frame.
PC1	 192.168.1.0/24 192.168.2.0/24 192.168.12.0/24 .10 PC1 Fa0 Fa0/6 Gig0/1 Gig0/0/0 Gig0/0/1 Gig0/0/1 Gig0/0/0 Gig0/1 SW1 Fa0/10 Fa0 PC2 .20 .1 R1 .1 .2 R2 .1 Fa0/1 Fa0/15 Fa0 PC3 .30 Fa0 Server .100 PDU Information at Device: PC1 OSI Model Inbound PDU Details Outbound PDU Details At Device: PC1 Source: Server Destination: PC1 In Layers Layer7 Layer6 Layer5 Layer4 Layer 3: IP Header Src. IP: 192.168.2.100, Dest. IP: 192.168.1.10 ICMP Message Type: 8 Layer 2: Ethernet II Header 000D.BDAC.B001 >> 00D0.58B3.6003 Layer 1: Port FastEthernet0 Out Layers Layer7 Layer6 Layer5 Layer4 Layer 3: IP Header Src. IP: 192.168.1.10, Dest. IP: 192.168.2.100 ICMP Message Type: 0 Layer 2: Ethernet II Header 00D0.58B3.6003 >> 000D.BDAC.B001 Layer 1: Port(s): FastEthernet0 1. FastEthernet0 receives the frame.

SW 1

```
SW1#show mac-address
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000d.bdac.b001    DYNAMIC     Gig0/1
SW1#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000d.bdac.b001    DYNAMIC     Gig0/1
SW1#
```

SW2

```
SW2>enable
SW2#show mac-address
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

SW2#
```